Neural-Symbolic NLP on a Small Transformer: Towards Neurosymbolic World Models for Manipulation

Ned Pham

***Abstract*—Neural networks trained by next-word prediction are strong at perception, but keep no explicit link to the rules behind a decision. Sheth, Roy and Gaur argue that safety-critical AI needs knowledge set as guardrails during training, not after it, and sort neurosymbolic methods into *lowering*, which compresses symbolic knowledge into the network, and *lifting*, which maps neural outputs to symbols for reasoning. On the lowering side, the semantic loss of Xu et al. turns a logical constraint into a training term. Ahmed et al. extend it to autoregressive models, where the exact probability of a constraint is #P-hard, and on GPT-2 pair it with word banning at decoding. A loss gives no test-time guarantee, and a mask leaves the model unchanged. This report asks what happens when one knowledge base drives both and a verifier lifts the output back to symbols, on a small transformer, and proposes the same split for neurosymbolic world models in robotic manipulation.**

***Index Terms*—Neuro-symbolic AI, lowering and lifting, semantic loss, constrained decoding, knowledge bases, world models, robotic manipulation.**

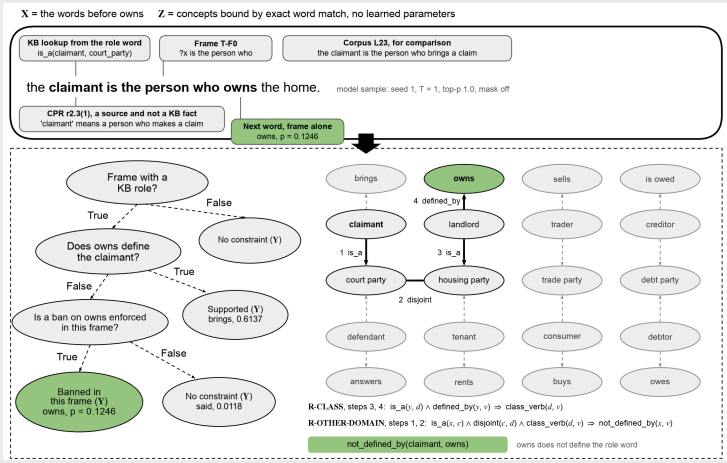


Fig. 1. **The core logic:** one sampled sentence traced through the civil-law KB. The frame matches, two rules derive *not_defined_by*(claimant, owns), so *owns* is banned and *brings* supported; $Y{=}\text{allowed}(\text{KB}, X)$. Seed 1, $T{=}1$, mask off; educational KB, not legal advice.

I. Introduction

Foundation models give robots broad priors but no guarantee that a step obeys a rule. Vision-language-action (VLA) policies above 90% success on standard LIBERO fall to 0.0% under simple perturbations [1]. On Towers of Hanoi, a PDDL-based neuro-symbolic system reached 95% against 34% for the best fine-tuned VLA [2].

The same gap appears in language at a scale small enough to inspect. In Sheth et al. [3], an intertwined pipeline earned 70% expert satisfaction against 47% for a federated LLM, and process knowledge [4], [5] asks systems to explain themselves by the expert's guideline. On GPT-2, a pseudo-semantic loss [6], [7] with word banning gave the lowest toxicity (9.20%, against 9.80% and 14.05% for either alone).

This report asks: *if one rule source is used in training, at decoding and to check the output, where does the constraint hold, and where does it fail?* The pairing is published [7]; this project is a version small enough to trace every step: a from-scratch TypeScript pipeline that climbs from a rule-based chatbot to a GPT-style transformer, then adds a knowledge base (KB) used as a semantic loss, a hard logit mask and a verifier (Fig. 2).

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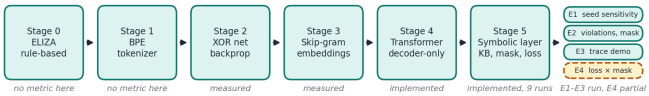


Fig. 2. Stages and status. Teal: done; amber: E4, held-out parts pending.

II. Method

A. Baselines before the symbolic layer

No KB, mask or semantic loss is used here. **XOR: same function, unrelated weights.** A single layer cannot learn XOR (loss 0.2845 → 0.2842); a 2→4→1 network can. At 5000 epochs all four seeds compute the same function with completely different hidden weights.

Skip-gram: the civil-law vocabulary, and a wrong answer. On England & Wales civil-dispute text (222 definitions, 34 stories), words that share sentences end up close (Fig. 3): buyer/customer 0.76, claimant/creditor 0.69. But none of five role analogies returns a role word (e.g. landlord − tenant + consumer gives *renters*, not *trader*). The model learned co-occurrence, not the relation.

Four seeds: the geometry agrees, the coordinates do not. On the earlier toy corpus, the correlation of pairwise geometry across seeds is 0.977–0.984, while the same word's vectors have cosine −0.28 to 0.09 (Fig. 4). The function is stable, the weights are not, and the same analogy error appears at every seed.

B. The transformer

Each token gets a query, a key and a value; attention is $\text{softmax}(QK^\top/\sqrt{d_k} + M)V$ with a causal mask M, so a

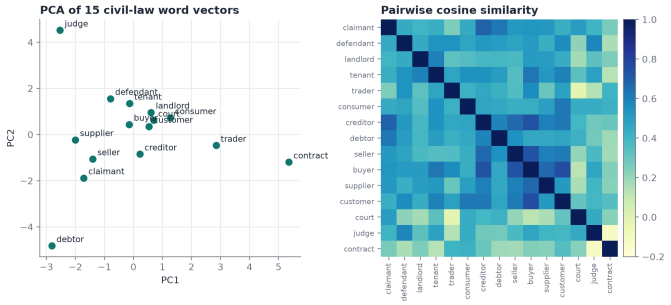


Fig. 3. PCA and cosine matrix of 15 civil-law word vectors, from the saved embeddings.

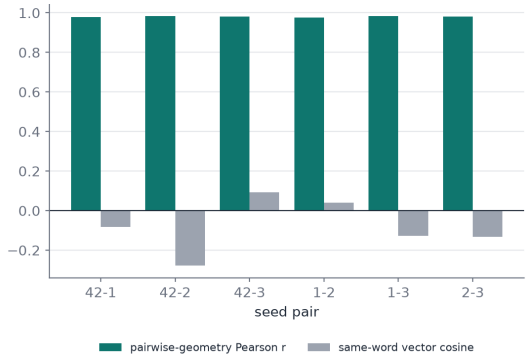


Fig. 4. Skip-gram seed agreement, toy corpus, six seed pairs: pairwise geometry agrees, same-word vectors do not.

token can only look left (Fig. 5). Several heads run in parallel, a feed-forward net refines each token, and blocks stack.

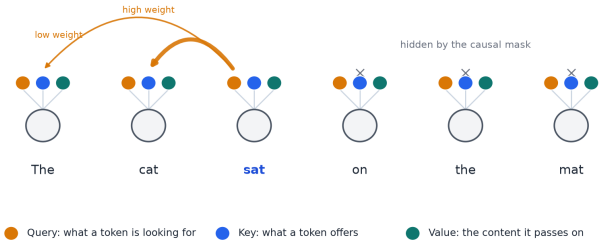


Fig. 5. Query, key and value. Schematic: arc width is an illustrative attention weight, not a measured one.

The block is pre-LayerNorm, as in GPT-2 [8] (Fig. 6): $x' = x + \mathrm{MHA}(\mathrm{LN}_1(x))$, $x'' = x' + \mathrm{FFN}(\mathrm{LN}_2(x'))$. Forward, backward, Adam and top-p sampling are written by hand and pass five gates, including a finite-difference gradient check. The model has 2 layers, d=32, 2 heads, feed-forward width 128 and context 32: 87,401 parameters with the 937-token whole-word BPE vocabulary.

The reasoning gap. The objective has no notion of truth: it rewards sounding like the corpus. Equal behaviour comes from unrelated weights, so attention maps show where the model looked, not a justification. The semantic loss has limits too: 69.89% of test predictions satisfied the constraint on grid paths in [6], and a wrong rule is trained in.

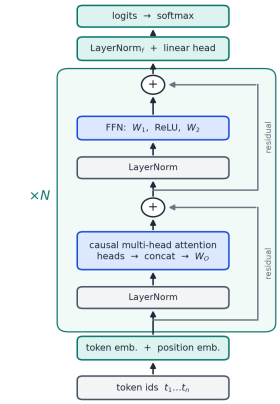


Fig. 6. The decoder-only block (pre-LN), implemented in `transformer.ts`. The shaded block repeats N times.

C. Mechanisms and the taxonomy

In the terms of Sheth et al. [3], *lowering* compresses knowledge into the network and *lifting* lifts its output into symbols (Fig. 7). In our mapping, the mask and soft bias are 1a-style lowering by analogy, the verifier and reasoning trace are decoupled lifting (2a), and the semantic loss moves toward intertwined integration (2b).

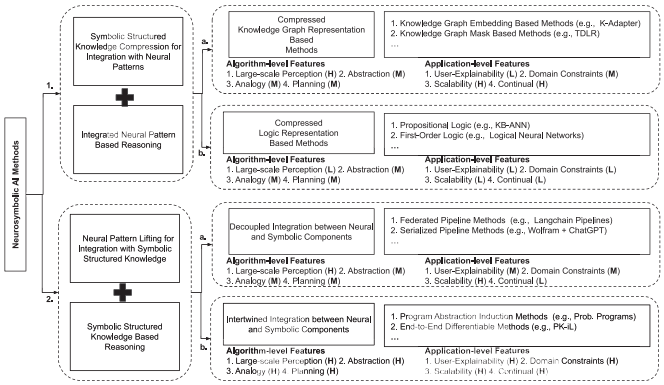


Fig. 7. Taxonomy of neurosymbolic AI methods, reproduced from [3], Fig. 1; L/M/H are its authors' ratings. Where this work sits is our mapping (text).

At decoding, the KB feeds a logits processor between the transformer and sampling, which records a reasoning trace (rules fired, mass moved), and a verifier checks the finished text (Fig. 8).

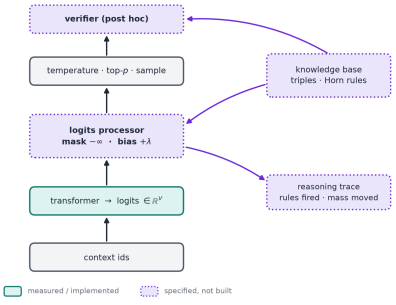


Fig. 8. Neuro-symbolic data flow at decoding. *Status note:* drawn before the symbolic layer was built; the parts marked "specified, not built" are now implemented.

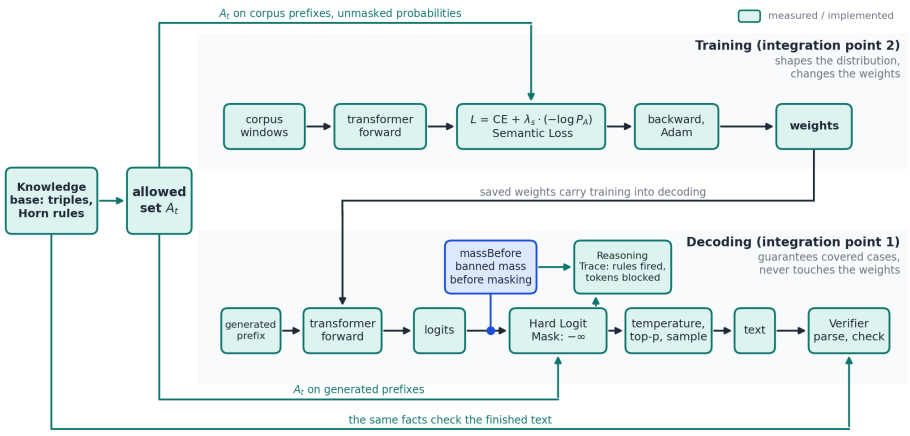


Fig. 9. One KB gives one allowed set, used by the semantic loss in training and by the hard mask at decoding; *massBefore* marks where the trained model meets the mask.

D. Pre-training: the baseline before the symbolic layer

The CE-only transformer trained 2000 epochs (seed 42) on all of the text. Its loss fell from 6.868 ($\approx \ln 937$) to 0.2489, but **this loss shows memorisation**: from epoch 1510 on, 100% of generated tokens sit inside runs of 8 or more tokens copied from the training text. Earlier, fluent samples stated definitions the corpus does not contain: "the claimant is the person who *owns* the home" (the corpus says *brings a claim*). Low loss is not correctness.

E. One knowledge base, two integration points

The two designs are not alternatives. A KB of triples plus Horn rules gives one allowed set A_t per position, and that set is used twice (Fig. 9): training shapes the distribution, and decoding guarantees the covered cases, as in [7].

The KB covers definitions in the corpus, e.g. with c the claimant and d the defendant, $\mathit{counterpart}(c, d) \wedge \mathit{defined_by}(d, \text{answers}) \Rightarrow \mathit{not_defined_by}(c, \text{answers})$. Forward chaining closes it once; a trigger turns a derived fact into banned token IDs when the context ends in a frame such as "the claimant is the person who", and every ban prints as a proof (Fig. 1: *owns*, p=0.1246, is banned).

Method 1: hard logit mask. Banned logits are set to $-\infty$ before temperature and top-p. This guarantees zero mass on covered banned tokens, but only as far as rules and triggers reach, and a wrong rule is enforced absolutely. It does not change the weights.

Method 2: semantic loss. $\mathcal{L} = \mathrm{CE} + \lambda_s(-\log P_A)$, with P_A the mass on allowed tokens [6], [7]. It depends on P_A alone, so it cannot see the choice among allowed tokens; it acts only where a rule fires (86 of 85,984 positions); and it is a soft penalty with no test-time guarantee.

Why the loss needs the mask. No finite λ_s removes all banned mass, so a correct rule still needs the mask; the loss shows in *massBefore* and in violations with the mask off.

Lifting. A verifier labels each generated claim *supported*, *contradicted* or *unknown*. A probe reads confident predictions as candidate facts (*confirmed*, *conflict*, *novel*), reported but never added to the KB. **Two trainers.** The baseline and the neural-symbolic model share corpus, tokenizer, model code and hyperparameters; only the loss differs, and λ_s=0 at seed 42 reproduces the baseline's weights exactly.

F. Protocol

Prompts are every role in every definition frame (8×3): 8 occur in the training text, and 16 put a role in a frame it never appears in ("the claimant is a business that"), the only held-out data. Each is sampled at seeds 0–49 for 20 tokens at two settings, mask off and on. A rule is violated when its banned word's mean p at its trigger is $\geq$0.02. All runs train 2000 epochs: one at λ_s=50 against the CE-only baseline (seed 42), and eight at $\lambda_s \in \{0, 2\}\times$ seeds 42, 1, 2, 3.

III. Results

A. One full-length run: λ_s=50 against CE only

At the 86 positions where a rule fires (Table I), CE only violates a rule at 1846 of 2001 epochs; the constrained model at 175, none after epoch 1746. The banned mass is lower at every logged epoch, and the semantic loss 73 times lower at epoch 2000; both curves rise and fall, peaking at 4.73e-4 (epoch 689) without the term and 2.26e-5 (epoch 585) with it (Fig. 10). Up to about epoch 600 the term also slows the allowed words; by epoch 1000 it is ahead. Training loss costs a little (0.2524 against 0.2489, on all text).

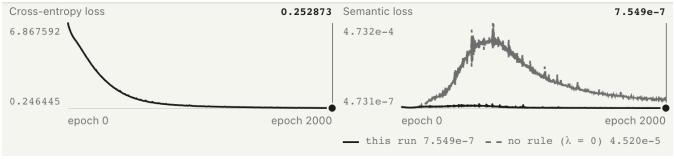


Fig. 10. The λ_s=50 run in the live demo: cross-entropy (left) and semantic loss (right) by epoch, λ_s=0 control dashed; values as logged in training (Table I reads the saved weights).

TABLE I
EPOCH 2000, SEED 42, THE 86 CONSTRAINED TRAINING POSITIONS.

Measure	CE only	CE + sem. loss
Semantic loss	4.58e-5	6.26e-7
Banned mass	0.0419	0.0006
p(next corpus word)	0.946	0.978
p(role's own verb)	0.809	0.841
Rules violated (of 48)	7	0

B. E2 and E3 at one training seed

With the mask on, 0 of 5,885 claims contradict the KB at this seed. With the mask off, the λ_s=50 model made no contradicted claim in 1,173 on the 8 training-text prompts (CE only: 1.2% at T=0.8, 4.8% at T=1). On the 16 untrained prompts it still contradicts the KB in 35–36% of claims (CE only 41–43%), mostly with the verb of the one role trained in that frame (315 of 430 are "sells" after "is a business that"): the frame, not the role, appears to drive it (untested). In E3, *massBefore* on banned verbs was 0.3252 (CE) against 0.0026.

C. E1 and E4: four training seeds

Table II compares λ_s=2 with 0. A paired sign-flip test over 4 seeds cannot go below p=0.125, so "at every seed" is not a significance claim.

TABLE II
SEEDS 42, 1, 2, 3, λ_s=0 VS 2, MASK OFF; RANGES OVER SEEDS OR SEED PAIRS.

Measure	λ_s=0	λ_s=2
Banned mass, 86 constrained pos.	0.024–0.049	0.008–0.016
Banned mass, 16 untrained prompts	0.45–0.73	0.43–0.71
Contradicted, untrained, T=0.8	35.4–47.9%	31.3–45.7%
JS of $(P_A, 1-P_A)$, mean	7.5e-3	2.5e-3
JS of q_A (among allowed), mean	0.0153	0.0149
Perplexity, no rule fires (train)	1.266–1.283	1.264–1.281

Seeds (E1). At λ_s=0, loss is similar but weights nearly unrelated; seeds agree on the next word at every training-text prompt, but at only 50–69% of held-out ones.

Semantic loss (E4). Banned mass at constrained positions is 2.6–4.3× lower at every seed; seeds disagree three times less on mass to allowed words, not *among* them, as Method 2 predicts. On untrained prompts the effect is small.

Mask coverage. With the mask on, 514 of 23,664 claims still contradict the KB, all "that *owed* money", which the mask bans only after "is": a coverage gap, not a model property.

IV. CONCLUSION

Where the constraint is trained, it holds, at a small cost in training loss. **Beyond it, the rule does not transfer**: on unseen frames the model follows the frame, not the KB. **A guarantee covers only what its rules cover**, as the "owed" gap shows. Limits: one small model, a hand-written KB, four seeds, no held-out text.

Towards neurosymbolic world models. Tokens become actions and states, and the allowed set becomes PDDL preconditions, effects and contact constraints such as slip bounds; the loss trains the world model, the mask rejects actions, and the verifier triggers recovery. *O1 (Years 1–2):* a multimodal model proposes PDDL action models under training and decoding constraints [6], [7], with semantic faithfulness checked by prover-based verification [9], [10] (GPT-4o's PDDL was 96.1% parseable but 24.8% correct [12]); it also tests whether world models follow plausibility over validity, like the frame effect above, and whether steering [13], [14] reduces it. *O2 (Years 2–3):* a slow symbolic layer sets subgoals and constraints; a fast controller rolls out an action-conditioned tactile model [15], [16] and checks constraints on predicted states, so a predicted violation triggers replanning before failure, compared with post-failure recovery [17], against VLA and LLM-planner baselines [1], [2]. The main risk is a wrong rule entering training and decoding at once, so every rule stays traceable to a proof.

Any LLM. As a *federated pipeline* [3], the access points of Fig. 9 become API levels: weights take the loss, logits the mask, text the verifier [9]–[11]. Designed, not built.

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